

A Two-Period Directed-Search Model with Wealth Heterogeneity

Equilibrium Definition and Numerical Solution

Simplified version of Repele (2026): no heterogeneity in job type x .
All jobs have common productivity y_{bar} , separation probability δ ,
retention $\Delta = 1 - \delta$, and vacancy cost ϕ .

Workers differ in initial liquid wealth $a_1 \sim G(a_1)$. Preferences:

$$u(c_1) + \beta E[u(c_2)], \quad u' > 0, \quad u'' < 0.$$

Savings satisfy $a_2 \geq a_{\text{bar}}$ and earn gross return R .

All workers start period 1 unemployed; unemployment income in period 2 is b .

A submarket is indexed by posted wage w . Let $n(w)$ be applicants and $v(w)$ vacancies. In an active submarket:

$$\theta(w) = v(w) / n(w).$$

Worker job-finding probability: $p(\theta)$, increasing in θ .

Vacancy-filling probability: $q(\theta) = p(\theta)/\theta$, decreasing in θ .

Period-2 values:

$$E_2(a_2, w) = u(R a_2 + w)$$

$$U_2(a_2) = u(R a_2 + b)$$

$$J(w) = y_{\text{bar}} - w.$$

Definition of equilibrium

A directed-search equilibrium is

$E = \{ a_2^*(a_1), w^*(a_1), \theta(w), n(w), v(w) \}$
such that:

1. WORKER OPTIMIZATION.

Taking the entire tightness schedule $\theta(w)$ as given, each worker solves

$$(a_2^*(a_1), w^*(a_1)) \text{ in } \arg \max_{\{a_2 \geq \bar{a}, w\}} \{ \\ u(R a_1 - a_2) \\ + \beta p(\theta(w)) \Delta u(R a_2 + w) \\ + \beta [1 - p(\theta(w)) \Delta] u(R a_2 + b) \}.$$

The probability of employment in period 2 is $p(\theta(w)) \Delta$.

2. FIRM FREE ENTRY.

Vacancy value in wage submarket w :

$$V(w) = -\phi + \beta \Delta q(\theta(w)) (\bar{y} - w).$$

Free entry / complementary slackness:

$$\begin{aligned} \phi &\geq \beta \Delta q(\theta(w)) (\bar{y} - w), \\ \theta(w) &\geq 0, \\ \theta(w) [\phi - \beta \Delta q(\theta(w)) (\bar{y} - w)] &= 0. \end{aligned}$$

Hence every active submarket satisfies

$$\phi = \beta \Delta q(\theta(w)) (\bar{y} - w).$$

3. WORKER-FLOW CONSISTENCY.

$$n(w) = \int 1_{\{w^*(a_1)=w\}} dG(a_1).$$

On a discrete wealth grid: $n_j = \sum_i \mu_i 1_{\{w_i^*=w_j\}}.$

4. VACANCY / TIGHTNESS CONSISTENCY.

$$v(w) = \theta(w) n(w) \quad \text{in every active submarket.}$$

5. MATCHING CONSISTENCY.

$$p_w = p(\theta(w)), \quad q_w = q(\theta(w)) = p(\theta(w))/\theta(w).$$

There is no asset-market-clearing condition because R is exogenous.

Why there is no fixed point over $\theta(w)$

Conditional on w , free entry determines $\theta(w)$ point by point:

$$q(\theta(w)) = \phi / [\beta \Delta (\bar{y} - w)].$$

Under CRS matching, firm profitability depends on n and v only through $\theta = v/n$. It does not depend on the absolute mass or composition of applicants. Thus the solution sequence is

$$w \rightarrow \theta(w) \rightarrow \text{worker choices} \rightarrow n(w) \rightarrow v(w) = \theta(w)n(w).$$

A genuine fixed point would arise if firm profitability depended on the endogenous composition of applicants (for example worker productivity types).

MATCHING-FUNCTION PARAMETERIZATION (Repele 2026, Appendix B1).

$$M(n, v) = n v (n^\gamma + v^\gamma)^{-1/\gamma}, \quad \gamma > 0.$$

With $\theta = v/n$:

$$\begin{aligned} p(\theta) &= \theta (1 + \theta^\gamma)^{-1/\gamma} \\ q(\theta) &= (1 + \theta^\gamma)^{-1/\gamma}. \end{aligned}$$

Define

$$q_{FE}(w) = \phi / [\beta \Delta (\bar{y} - w)].$$

For $0 < q_{FE}(w) < 1$, free entry can be inverted analytically:

$$\theta(w) = (q_{FE}(w)^{-\gamma} - 1)^{1/\gamma}.$$

Then $p(w) = p(\theta(w))$. If $q_{FE}(w) \geq 1$, the submarket is inactive.

Repele's quantitative model uses $\gamma = 0.38$. Her quantitative model also uses CRRA $\sigma = 2$; both are natural benchmark values for a numerical example.

Numerical parameterization and grids

Normalize $y_{\text{bar}}=1$. Choose β and R consistently with the period length.
Choose common δ and set $\Delta=1-\delta$. A simple benefit rule is
$$b = \rho_b y_{\text{bar}}.$$

CALIBRATING VACANCY COST.

Pick reference wage w_{ref} and target job-finding probability p_{ref} .

For the matching function above:

$$\begin{aligned}\theta_{\text{ref}} &= [p_{\text{ref}}^{\gamma}/(1-p_{\text{ref}}^{\gamma})]^{(1/\gamma)} \\ q_{\text{ref}} &= (1-p_{\text{ref}}^{\gamma})^{(1/\gamma)} \\ \phi &= \beta \Delta q_{\text{ref}} (y_{\text{bar}}-w_{\text{ref}}).\end{aligned}$$

WAGE GRID.

Active submarkets require $q_{\text{FE}}(w) \leq 1$, hence

$$w \leq y_{\text{bar}} - \phi/(\beta \Delta).$$

With no nonpecuniary value of employment, $w \leq b$ is dominated. Use

$$w_{\text{min}} = b + \epsilon_w$$

$$w_{\text{max}} = y_{\text{bar}} - \phi/(\beta \Delta) - \epsilon_w.$$

A uniform grid with about 200-500 points is adequate.

Because $\theta(w)$ can vary rapidly near w_{max} , an alternative is a grid uniform in θ or p , recovering wages from

$$w(\theta) = y_{\text{bar}} - \phi/[\beta \Delta q(\theta)].$$

WEALTH GRID.

With $a_{\text{bar}}=0$, use a_i in $[0, a_{\text{max}}]$, concentrated near zero:

$$a_i = a_{\text{max}} [(i-1)/(N_a-1)]^{\kappa}, \quad \kappa > 1.$$

For illustration, $\kappa=2$ and N_a around 200 are adequate.

Attach masses μ_i from a parametric or empirical liquid-wealth distribution.

Solution algorithm

Step 1. Choose wage grid $\{w_j\}$, wealth grid $\{a1_i\}$, and masses $\{\mu_i\}$.

Step 2. For each wage w_j compute

$$q_{FE_j} = \phi / [\beta \Delta (y_{\bar{}} - w_j)].$$

If $0 < q_{FE_j} < 1$, compute θ_j from the analytical inversion and $p_j = p(\theta_j)$. Otherwise mark the submarket inactive.

Step 3. For each pair $(a1_i, w_j)$, solve the one-dimensional savings problem

over $a2$:

$$\begin{aligned} \max_{a2} & u(R a1_i - a2) \\ & + \beta p_j \Delta u(R a2 + w_j) \\ & + \beta (1 - p_j \Delta) u(R a2 + b). \end{aligned}$$

A convenient feasible interval is $[a_{\bar{}}, R a1_i - \epsilon_c]$.

Step 4. For each wealth type, choose the wage index $j^*(i)$ giving the

largest optimized value. Store w_i^* , θ_i^* , p_i^* , and $a2_i^*$.

Step 5. Count applicants:

$$n_j = \sum_i \mu_i 1\{j^*(i) = j\}.$$

This is the mass of period-1 searchers in the submarket, not an unemployment rate.

Step 6. Set vacancies in each active submarket:

$$v_j = \theta_j n_j.$$

If $n_j = 0$, set $v_j = 0$.

Step 7. Verify free entry, $\theta_j = v_j / n_j$, and worker optimality.

Output: $a1 \rightarrow (a2^*(a1), w^*(a1), \theta^*(a1), p^*(a1))$, plus $n(w), v(w)$.

REFERENCE

Repele, Amalia (2026), Wealth Sorting and Cyclical Employment Risk, Job Market Paper, International Institute for Economic Studies, Stockholm University, July 2026.